

Wash lending in vault networks: funding, incentives and loss transmission

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Abstract

A vault can finance borrowing against its own shares, allowing borrowers to redeposit the proceeds and recover part of their interest payments through share ownership. We study the resulting wash-lending loop in a closed model with a fixed allocation between an external strategy and a lending market. Consolidation shows that the loop can deploy idle cash but cannot expand external investment beyond initial equity. Borrowing redistributes the resulting income, with different incentives for price-taking and strategic participants. We derive the funding bound, the return decomposition and local liquidation thresholds, then examine how losses in self-referential collateral can feed back into vault value. The analysis distinguishes the priority of a secured loan from the rights of passive vault holders, and market funding attribution from verified circular borrowing. These distinctions provide a basis for evaluating self-lending structures without treating gross balance sheet growth, interest rebates and loss protection as equivalent.

1 Introduction

Consider a vault that promises access to an external investment strategy while retaining part of its assets in a lending market. Suppose that market accepts the vault's own shares as collateral. A holder can then borrow from the vault, deposit the proceeds back into it, and pledge the newly issued shares. The vault grows, its external investment can grow, and the borrower receives a larger share of its income. Yet no new investor has supplied capital to the system. What has the loop financed, and who bears the obligations it creates?

We call this arrangement *wash lending*: borrowing funded by an issuer or vault against claims on itself, with the proceeds returning to the same balance sheet. The term describes a financing structure; it does not by itself imply fictitious transactions or misconduct. The economic issue is the overlap between the roles of borrower, shareholder and lender. Interest paid by a borrower becomes vault income, part of which returns to that borrower through its shares. A contractual interest rate is therefore not enough to identify the borrower's net payment to other participants.

1.1 The question and the contribution

The central question is how a loop can improve individual returns without creating aggregate income from its internal payments. Our answer starts with the use of cash. Under a fixed allocation rule, redeposited borrowing directs a fraction of idle lending liquidity into the external strategy. That additional deployment can generate income. Interest then distributes the income between participants according to their debt and share ownership. The distinction matters: an increase in a holder's return can reflect additional external investment, a transfer from another holder, or both.

The same accounting imposes a limit. If all borrowing is funded by the vault, external investment cannot exceed the participants' initial equity. A collateral rule allowing high leverage on an individual position does not supply the cash required for that leverage across the whole vault. Moreover, receiving an interest rebate does not establish an incentive to borrow. A participant must compare leverage with holding unlevered shares, which receive the same base vault yield. A large holder also changes the rebate by changing its ownership share; its decision differs from that of a price taker.

Risk introduces a further distinction. Borrower equity protects the secured loan against a local collateral loss. It does not give passive holders of the same vault share class priority in the vault's assets. When those shares also secure loans on the vault's balance sheet, a loss in external backing can affect both share value and loan recovery. The paper develops these accounting and incentive results and uses them to identify what a reported self-lending exposure can establish. It does not estimate the probability of a run or claim that circular funding alone determines its outcome.

1.2 Approach, scope and relation to the literature

We begin with a numerical example and then consolidate a closed network consisting of a vault and its participants. All borrowing proceeds are redeposited; the vault owns all lending claims; share income is distributed pro rata; idle cash earns zero. The external strategy's return and the allocation rule are specified rather than derived from asset-market equilibrium. This deliberately narrow setting makes the sources and recipients of income explicit. We subsequently distinguish price-taking leverage choices from a strategic borrower's choice, and use an exogenous collateral markdown to separate liquidation from equity exhaustion. Opening the funding perimeter to outside lenders is treated in an appendix.

The analysis relates to the financial-network perspective of Eisenberg and Noe [1], where interdependent obligations must be valued consistently with limited liability and debt priority. Our self-referential NAV equation is a stylized valuation condition, not an application of their clearing theorem: vault shares and their redemption introduce claims whose settlement requires additional rules. It also relates to the collateral-equilibrium literature reviewed by Fostel and Geanakoplos [2], which studies feedback between leverage and asset values. Here collateral limits are given, and the focus is the funding and income feedback created when the lender holds claims secured by its own shares.

Section 2 follows the cash through a loop and a redemption. Sections 3 and 4 establish what is conserved and how income is shared. Section 5 asks when those payments induce leverage; Section 6 follows the obligations under stress. Section 7 turns the model into an

interpretation of market evidence. The epilogue returns to the design question raised by the opening example: what does the loop enable while it is running, and what must remain possible when a holder wants to leave?

2 What wash lending changes: a motivating example

A vault V accepts asset S and issues shares X . It allocates half its assets to an external strategy L earning $s = 8\%$, and half to a market $\mathcal{M}$ lending S against X at $r = 3\%$. The vault is the sole lender and the participants below are its only shareholders. Unborrowed cash earns zero. Rates are annualized simple income rates, held constant for each comparison; the calculations are not compounded APYs and exclude fees and losses.

Transactions occur before income accrues, with initial share NAV normalized to $1S$. All shares minted by a looping borrower are also pledged as collateral. The final position can be reached in successive borrow, deposit and pledge steps, subject to liquidity and collateral limits at every step. Figure 1 shows the flows between the three parties.

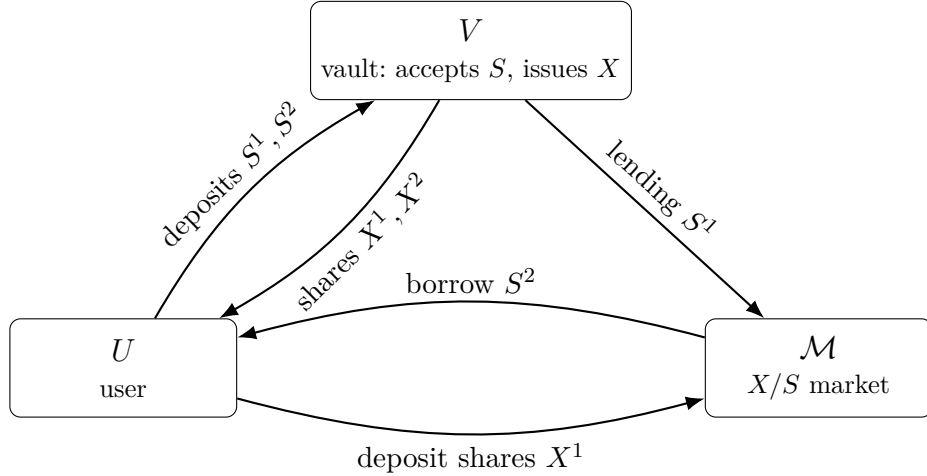


Figure 1: Flows in the motivating example. The user U deposits S^1 and S^2 into the vault V and receives shares X^1 and X^2 . The vault lends S^1 to the market $\mathcal{M}$; the user pledges X^1 there and borrows S^2 , which is redeposited in V .

2.1 Initial deposits

P_1 and P_2 each deposit $100S$ and receive $100X$. The vault holds $100S$ in L and $100S$ idle in $\mathcal{M}$. Its annual income is $100 \times 8\% = 8S$, so each participant earns $4S$.

2.2 Borrowing and redepositing

P_2 pledges its shares, borrows $60S$, deposits the proceeds in V , and pledges the additional $60X$. The resulting positions are

$$P_1 = 100X, \quad P_2 = 160X - 60S, \quad T = 260S.$$

The vault has $130S$ in L and a $130S$ lending position consisting of $60S$ of loans plus $70S$ of cash. Its income and the participants' net income are

$$\begin{aligned} I_V &= 130(0.08) + 60(0.03) = 12.2, \\ I_1 &= \frac{100}{260}(12.2) \approx 4.69, \\ I_2 &= \frac{160}{260}(12.2) - 1.8 \approx 5.71. \end{aligned}$$

The $1.8S$ interest payment is internal: $I_1 + I_2 = 10.4S$, exactly the income of L . Of its interest payment, P_2 recovers $160/260$; the net payment to P_1 is $1.8(100/260) \approx 0.692S$, equivalent to a rate of 1.154% on the $60S$ debt.

P_2 's debt/equity multiplier is $m = 0.6$, while its lending-market LTV is $60/160 = 37.5\%$. Using only the original $100X$ as pledged collateral would instead give 60% LTV; that is a different position. The income gain comes from moving $30S$ of initially idle cash into L . Allocating the initial $200S$ entirely to L would have earned $16S$, exceeding the loop's combined income, but would change the vault's allocation and liquidity profile.

2.3 Redemption changes ownership

Suppose P_1 redeems $100X$ immediately, before income accrues. Assume L is redeemable at par: V withdraws $50S$ from each venue, leaving $80S$ in L and $80S$ in $\mathcal{M}$, of which $60S$ is lent. P_2 still has $160X - 60S$ and now owns all vault shares. Its net income becomes

$$I_2 = 80(0.08) + 60(0.03) - 60(0.03) = 6.4S.$$

It recovers all interest paid to the vault. LTV is unchanged, but market utilization rises from $60/130$ to $60/80$. An actual rate model could therefore change the quoted borrow rate. Complete rebate does not make pledged shares freely redeemable or make loans liquid.

The example separates the benefit of deploying cash from the benefit of receiving interest paid by someone else. It also reveals their liquidity cost: the external strategy holds more assets and the lending market retains less cash. To determine how far this process can go, we next consolidate the balance sheets.

3 How the loop works: accounting and funding

3.1 Participants, claims and the consolidation boundary

Let $e_i > 0$ be participant i 's initial equity, $y_i = m_i e_i$ its debt, and $h_i = e_i + y_i$ its share value at the initial NAV. Include passive holders as $m_i = 0$. Write

$$E = \sum_i e_i, \quad Y = \sum_i y_i, \quad T = E + Y, \quad \varphi = \frac{Y}{T}.$$

The vault allocates fraction $0 < \varepsilon < 1$ to L , with return $s \geq 0$, and $1 - \varepsilon$ to $\mathcal{M}$, at borrow rate $r \geq 0$. All market debt is owed by these participants, all proceeds are redeposited, and

the vault owns all lending claims. These closure assumptions are needed for the identities below. At a later NAV q , a deposit of dS mints d/q shares; nominal share counts must not be confused with asset values.

The vault's assets are

$$\underbrace{\varepsilon T}_L + \underbrace{Y}_{\text{loans}} + \underbrace{C}_{\text{idle cash}} = T, \quad C = (1 - \varepsilon)T - Y = E - \varepsilon T.$$

Pledged X is security for the loans, not an additional vault asset. Consolidating the vault and its shareholders cancels the internal loan and share claims and leaves $\varepsilon T + C = E$ of external assets.

3.2 The funding bound

Proposition 1 (Funding and external investment). *Under the closed-network assumptions above, every feasible position satisfies*

$$Y \leq (1 - \varepsilon)(E + Y) \iff \frac{Y}{E} \leq \frac{1 - \varepsilon}{\varepsilon}. \quad (1)$$

External investment obeys $\varepsilon T \leq E$, with equality at full market utilization. For $s \geq 0$, external income is at most sE .

Proof. The outstanding loans cannot exceed the vault's lending allocation: $Y \leq (1 - \varepsilon)T$. Substituting $T = E + Y$ yields (1). Equivalently, idle cash $C = E - \varepsilon T$ must be nonnegative. At full utilization $C = 0$, so $\varepsilon T = E$. Multiplication by $s \geq 0$ gives the income bound. $\square$

Define utilization

$$u = \frac{Y}{(1 - \varepsilon)(E + Y)}, \quad \varphi = (1 - \varepsilon)u.$$

Full utilization fixes $Y = (1 - \varepsilon)E/\varepsilon$ and $\varepsilon T = E$. Thus looping can deploy all initial equity into L , but cannot increase the external allocation beyond E in this closed model. Combined income is at most sE . Individual profits can still differ because interest redistributes that income.

3.3 Collateral capacity and redemption capacity

Funding and collateral are separate constraints. With liquidation LTV ℓ_i , a recursive position must satisfy

$$\frac{m_i}{1 + m_i} \leq \ell_i, \quad m_i \leq \frac{\ell_i}{1 - \ell_i}.$$

A practical maximum M_i can be lower to allow a liquidation buffer. A feasible terminal balance sheet does not itself prove that a particular transaction sequence has enough liquidity to reach it.

A redemption of wS that preserves the allocation fractions requires $(1 - \varepsilon)w \leq C$ if debt remains outstanding. It also requires εw to be withdrawable from L . Without that allocation constraint, liquid L assets may fund more redemptions, but the vault's portfolio then changes.

The funding bound settles what the loop can invest. It leaves open who benefits from that investment, since a loan is an asset of the vault and a liability of a participant who also owns vault shares. The next step is to follow both sides of each interest payment.

4 How the loop pays: income and ownership

4.1 Conservation and distribution of income

Proposition 2 (Return decomposition). *Vault income is $I_V = \varepsilon sT + rY$. The return per unit of vault shares and participant i 's net return on equity are*

$$c_v = \varepsilon s + r\varphi, \quad (2)$$

$$A_i = (1 + m_i)c_v - rm_i = c_v + m_i\Delta_v, \quad \Delta_v = \varepsilon s - r(1 - \varphi). \quad (3)$$

Consequently,

$$\sum_i e_i A_i = \varepsilon sT. \quad (4)$$

Proof. Participant i receives $h_i I_V / T$ and pays ry_i . Dividing the difference by e_i gives (3). Summing net income over participants cancels the rY paid to and received from the vault, leaving (4). $\square$

A passive holder earns c_v , including borrow-market income. It does not earn only εs .

4.2 A borrower's own payment and marginal carry

Participant i owns fraction $\alpha_i = h_i / T$ of the shares. Of its own payment ry_i , it recovers $\alpha_i ry_i$, so the net payment to other holders is $(1 - \alpha_i)ry_i$. It also receives $\alpha_i r(Y - y_i)$ from other borrowers. Hence

$$e_i A_i = \varepsilon s h_i - r(1 - \alpha_i)y_i + \alpha_i r(Y - y_i).$$

For a sole borrower with passive equity x_1 and own equity x_2 , this reduces to

$$T = x_1 + x_2 + y, \quad b_{\text{eff}} = r \frac{x_1}{T}, \quad I_2 = \varepsilon s(x_2 + y) - b_{\text{eff}}y.$$

Relative to no loop, its gain is $y(\varepsilon s - rx_1/T)$; the passive holder gains rx_1y/T . A concentrated borrower can therefore benefit even when the quoted rate exceeds s , subject to funding and collateral feasibility.

By contrast, $r(1 - \varphi)$ in (3) is the financing term in the *price-taking incremental carry*. It is not generally the net rate on a particular borrower's own interest bill. The ownership share α_i and the aggregate debt fraction φ are distinct.

The distinction between an ownership rebate and incremental carry is the bridge from accounting to behavior. The payment identity holds for any feasible position. To predict a position, however, we must specify what the participant can change and which quantities it takes as given.

5 From the rebate to leverage choice

5.1 The price-taking benchmark

A participant treating r and φ as fixed chooses $m_i \in [0, M_i]$ to maximize (3). The best response is

$$m_i \in \begin{cases} \{M_i\}, & \Delta_v > 0, \\ [0, M_i], & \Delta_v = 0, \\ \{0\}, & \Delta_v < 0. \end{cases}$$

If a_i is an outside-option return, participation additionally requires

$$\max\{c_v, c_v + M_i \Delta_v\} \geq a_i.$$

The test $A_i(M_i) \geq a_i$ alone is insufficient: when $\Delta_v < 0$, holding unlevered shares pays more than borrowing. If participants withdraw equity to take their outside option, E must also be determined in equilibrium. Holding E fixed instead describes a population whose equity remains in the vault.

At fixed Y , a participant with $m_i < Y/E$ receives net interest:

$$\left. \frac{\partial A_i}{\partial r} \right|_{Y, m_i} = \frac{Y - m_i E}{E + Y} > 0.$$

This is a wealth-transfer effect. It does not imply upward-sloping optimal borrowing demand: at the same fixed aggregate state, $\partial \Delta_v / \partial r = -(1 - \varphi) < 0$. A higher rate can make share ownership more attractive while making additional leverage less attractive.

5.2 Closing the model with supply and rates

For a specified utilization-based rate rule $r = r(u)$, fixed equity, and no participation exit, the benchmark equilibrium solves

$$Y = \sum_i e_i m_i, \quad u = \frac{Y}{(1 - \varepsilon)(E + Y)}, \quad m_i \in \text{BR}_i(r(u), \varphi), \quad 0 \leq u \leq 1.$$

This is the endogenous-supply formulation. Imposing an unrelated fixed supply $\bar{S}$ and $Y = \bar{S}$ would conflict with the vault allocation unless $\bar{S} = (1 - \varepsilon)E/\varepsilon$, or the allocation rule were explicitly changed.

At fixed r , more aggregate debt raises each fixed- m_i return:

$$\left. \frac{\partial A_i}{\partial Y} \right|_{r, m_i} = (1 + m_i)r \frac{E}{(E + Y)^2} \geq 0.$$

But with a differentiable equilibrium rate schedule $r(Y)$,

$$\begin{aligned} \frac{d\Delta_v}{dY} &= \frac{E}{(E + Y)^2} [r(Y) - (E + Y)r'(Y)], \\ \frac{dA_i}{dY} &= (1 + m_i)r \frac{E}{(E + Y)^2} + [(1 + m_i)\varphi - m_i]r'(Y), \end{aligned}$$

where m_i is held fixed. Rising utilization can offset the rebate. Multiple equilibria require a demonstrated fixed-point structure; they do not follow from the rebate identity alone.

Corollary 3 (A limit to price-taking carry). *For $r \geq 0$, a feasible closed-network position satisfies*

$$\Delta_v \leq \varepsilon(s - r).$$

Consequently $r \geq s$ rules out strictly positive price-taking carry.

Proof. Equation (1) implies $1 - \varphi = E/T \geq \varepsilon$. Substitute this inequality into $\Delta_v = \varepsilon s - r(1 - \varphi)$. $\square$

This result links the behavioral calculation back to the funding bound. A rebate cannot be increased independently of the cash needed to finance it. The conclusion concerns incremental leverage at a given aggregate state; a strategic borrower's finite change in ownership requires a different comparison.

5.3 The strategic borrower

A large borrower changes its ownership share and the rebate by looping. Hold all initial equities and the other borrowers' debt Y_{-i} fixed, and let y be borrower i 's choice. With $T = E + Y_{-i} + y$ and $H_{-i} = E - e_i + Y_{-i}$, its income is

$$R_i(y) = \varepsilon s(e_i + y) + r \left[\frac{(e_i + y)(Y_{-i} + y)}{T} - y \right].$$

At a fixed quoted rate,

$$R'_i(y) = \varepsilon s - \frac{rEH_{-i}}{T^2}, \quad R''_i(y) = \frac{2rEH_{-i}}{T^3} \geq 0.$$

The objective is convex, rather than linear. On a feasible interval a maximum occurs at an endpoint, but the decision must compare total income at those endpoints; the atomistic carry test is not its rule. If the borrower also moves $r(Y)$, the first derivative gains the term

$$r'(Y) [(e_i + y)\varphi - y],$$

and convexity need not survive. Entry or exit can lower interest costs and reduce rebates simultaneously, so their combined effect has no universal sign.

A return calculation can therefore explain why a loop is chosen without establishing that it can be unwound. The remaining question concerns the claims themselves: what happens when the collateral behind a loan is also a claim on the lender's assets?

6 When the loop is stressed: losses and recovery

6.1 The local secured-loan calculation

Consider a position with equity e , collateral $(1+m)e$, and debt me , all valued in the loan asset. Let collateral fall by fraction $\delta \in [0, 1]$, holding debt fixed. Without intervening liquidation,

remaining borrower equity, borrower loss and lender shortfall are

$$\begin{aligned} Q(\delta) &= e [1 - (1 + m)\delta]_+, \\ L_J(\delta) &= e \min\{(1 + m)\delta, 1\}, \\ L_D(\delta) &= e [(1 + m)\delta - 1]_+. \end{aligned}$$

They satisfy $L_J + L_D = \delta(1 + m)e$. The lender shortfall begins *after* the borrower has lost its equity; it must not be charged against that equity a second time.

For $m > 0$, the liquidation trigger and equity-wipeout threshold are

$$\boxed{\delta_{\text{liq}} = 1 - \frac{m}{\ell(1 + m)}, \quad \delta_A = \frac{1}{1 + m}.} \quad (5)$$

An initially healthy position has $\delta_{\text{liq}} \geq 0$. At the maximum multiplier $m = \ell/(1 - \ell)$ it has no initial liquidation buffer. For $0 < \ell < 1$, liquidation is triggered before equity wipeout.

m	ℓ	δ_{liq}	δ_A
4	0.830	3.61%	20.00%
6	0.915	6.32%	14.29%
15	0.945	0.79%	6.25%

These are illustrative parameters, not claims about current markets. Liquidation incentives, slippage, oracle delays and incomplete execution change actual recoveries. For example, if liquidation of all collateral repays only a fraction $\eta \leq 1$ of its marked value, the residual shortfall is $[me - \eta(1 - \delta)(1 + m)e]_+$; this is a recovery scenario, not the no-liquidation equity calculation above. Morpho's published bad-debt description explicitly distinguishes the LLTV trigger from collateral exhaustion after the liquidation incentive [3].

With heterogeneous positions and separate collateral accounts, aggregate lender shortfall under the same exogenous shock is

$$L_D^{\text{agg}}(\delta) = \sum_{i:m_i>0} e_i [(1 + m_i)\delta - 1]_+.$$

The first positive shortfall appears just above $\min_{i:m_i>0} 1/(1 + m_i)$. One borrower's unused equity does not protect another borrower's lender unless there is an explicit pooled guarantee. There is no additional test equating summed shortfalls to all borrowers' initial equity.

6.2 Debt priority and the common share class

The loan claim has priority over the borrower's residual collateral value. But passive and looping holders own the same vault share class in this model. Passive holders have no contractual priority over looping holders in redeeming V , and a loss in L reduces the value of every share immediately. Loan losses recognized by the vault also reduce the common NAV. The secured loan has a junior equity buffer; the whole vault is not thereby a CDO with passive holders as seniors.

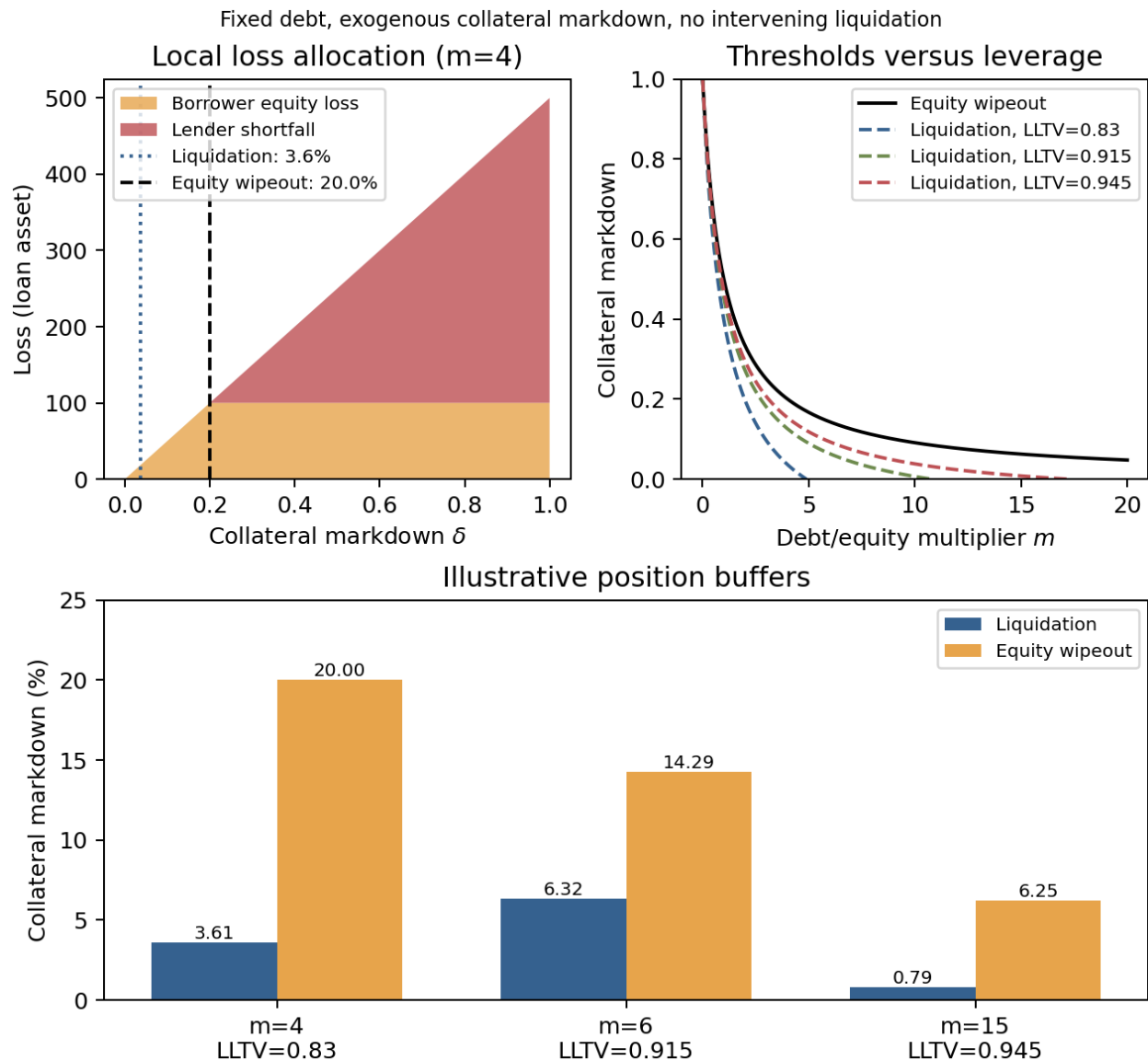


Figure 2: Local secured-loan losses and thresholds, generated by `wash_tranche.py`. The loss calculation assumes an exogenous collateral markdown, fixed debt and no intervening liquidation. The liquidation line is a reference trigger. Curves are restricted to initially feasible leverage. These are not a consolidated vault-loss waterfall or a calibrated default forecast.

This matters because the collateral is itself a claim on the lender. If B is the post-shock value of the vault’s external assets including cash, q its share NAV, and n_i borrower i ’s pledged share count, a stylized simultaneous mark is

$$qT = B + \sum_i \min\{y_i, qn_i\},$$

where initial outstanding shares are T , and liquidation costs and share burns are omitted. If all loans remain recoverable, B falling by λ lowers NAV by λ/T at once. If $qn_i < y_i$, loan write-downs feed back into NAV. Recoverability depends on who can buy or redeem seized shares and with what external liquidity. The formula is a consistency condition under its marking assumptions, not an execution or settlement model. It shows why one cannot add pledged share values to vault assets as independent backing.

6.3 Price risk and the time to exit

For ETH-denominated loans against ETH-denominated vault shares, the relevant collateral return is the share/ETH basis. At the initial mark, $(1 + m)e - me = e$ leaves net ETH exposure on equity, but leverage still amplifies losses in that basis. Small observed day-to-day volatility does not establish safety against slashing, contract loss or a discontinuous discount on redemption.

If a zero-drift arithmetic return were Gaussian with annualized volatility σ , its terminal loss probability at horizon τ would be

$$\Pr(\delta_\tau > \delta_A) = \Phi\left(-\frac{\delta_A}{\sigma\sqrt{\tau}}\right).$$

This is not the probability of crossing the threshold at any time before τ . Under a driftless Brownian model that first-passage probability is $2\Phi(-\delta_A/(\sigma\sqrt{\tau}))$ for a positive constant barrier. Neither expression captures jumps, endogenous NAV feedback or failed liquidations. There is no probabilistic claim in Figure 2.

Stress analysis should distinguish oracle detection, liquidation execution, collateral sale or redemption, and lender withdrawal. The exit horizon depends on available cash, repayment flows and competing redemptions. It cannot be inferred from collateral volatility alone. Different shocks enter this sequence at different points. A fall in external income reduces carry; a loss of backing reduces NAV; issuance without corresponding assets dilutes claims; a withdrawal restriction changes the set of feasible exits. Under the price-taking assumptions, $\varepsilon s = 0$ and $r > 0$ imply negative incremental carry. That income statement alone determines neither the speed of deleveraging nor the associated capital loss.

This distinction sets the boundary of the model’s risk conclusions. Local leverage identifies a collateral buffer. Ownership identifies who receives income and shares recognized losses. Settlement and redemption rules determine whether collateral value can become cash in time. A market application must supply all of these inputs.

7 From the model to market evidence

The closed model is useful because every payment has an identified recipient and every loan has an identified source. Public market snapshots often provide a different object: outstanding borrowing against an issuer’s tokens and the issuer’s share of lending supply. To use that evidence, we must distinguish a funding attribution from the closed loop defined in Section 1.

7.1 What a funding attribution measures

For lender share α_k of market k and gross debt B_k against the relevant collateral, its pro-rata attribution is

$$F = \sum_k \alpha_k B_k.$$

Unlike $\varphi = Y/T$, α_k is a *share of market supply*. It is neither a borrower’s share of vault income nor the fraction of vault backing financed through loops.

Appendix C retains an attributed historical snapshot as an illustration of this calculation. Its reported funding and backing amounts produce a ratio near 30%. The role of the example is methodological: neither that ratio nor the accounting model provides a verified estimate of an issuer’s circular borrowing without evidence about the use of loan proceeds.

7.2 Tracing funds, income and claims

A loan against an issuer’s token need not finance a redeposit into that issuer. Proceeds may fund another asset or leave the system. Establishing $T = E + Y$ for this case therefore requires tracing borrowing proceeds, mints, redemptions and the relevant consolidation perimeter. Pro-rata funding attribution alone does not establish that identity or quantify TVL double counting.

The income mapping also requires care. USDe, sUSDe and maturity-specific principal tokens are not interchangeable pro-rata claims on all reserve income. Ethena documents discretionary reward distributions into sUSDe and states that deposited USDe is not itself lent out by the staking contract. Its reserve-fund and reward mechanics also prevent a direct identification of negative protocol income with negative staking rewards [4]. A case-specific model must identify which legal or contractual entity lends reserves, who receives that interest, and how it reaches each collateral token. The remaining backing cannot simply be assumed to be invested in a single basis strategy.

Even $\alpha_k = 100\%$ means only that the attributed lender supplies all liquidity in that market. A borrower does not recover all interest unless it also receives all the corresponding distributed income. Market lender concentration alone does not determine leverage, liquidation order, or the size of the borrower’s rebate.

7.3 Observable implications and limits

The useful follow-up to a snapshot is to measure the flow of borrowed funds, income pass-through, actual leverage and LLTV by position, oracle and redemption rules, and stressed exit liquidity. Those inputs connect a funding attribution to a model of returns and losses.

The model suggests concrete accounting checks. Where the closed-network assumptions hold, shareholder income net of borrowing costs must equal external strategy income. At fixed initial equity and allocation, additional redeposited debt must reduce idle cash by ε per unit borrowed. A rise in lending-market utilization without a corresponding change in the measured debt fraction would signal that the assumed funding perimeter or allocation has changed. These checks test the mapping of a system to the model before any behavioral or risk interpretation is attempted.

The behavioral claims require additional evidence. An estimated response of borrowing to rates must distinguish changes in the outside option, income distribution, ownership concentration and loan supply. Observed positive net interest income for a low-leverage holder is not evidence that higher rates caused that holder to borrow more. Likewise, testing loss transmission requires realized recovery prices and redemption flows, rather than collateral marks alone.

The present analysis holds the external strategy, loan denomination and allocation rule fixed. Fees, rewards paid from outside the modeled system, asset-price effects, heterogeneous claim classes and strategic withdrawals can change the conclusions. A dynamic extension would need to specify the timing of interest accrual, NAV updates, margin calls and cash settlement. The identities derived here provide consistency conditions for that extension, not a substitute for it.

8 Epilogue: what remains when the loop unwinds

Return to the initial vault. A holder borrows $60S$, redeposits it, and increases the external investment by $30S$. Both participants receive more income than before the loop. The improvement is real within the model: previously idle cash has been put to work. The interest paid along the way determines how that income is shared. Consolidation makes the source of the improvement visible without erasing the obligations that make it possible.

Those obligations matter when the passive holder leaves. Its exit changes the ownership of the interest stream, consumes cash and raises utilization even though the remaining borrower's LTV is unchanged. The remaining holder may recover every interest payment and still face a binding constraint on redemption. The ability to receive the income on a claim is different from the ability to turn that claim into the asset needed to repay a loan.

For a vault designer, this gives a practical interpretation of the funding bound. An allocation to lending is also a reserve of potential withdrawal liquidity, to the extent that it remains unborrowed. Looping can convert that cash into external investment, while changing who depends on future repayments or liquidations. Evaluating the conversion requires the external strategy's return and liquidity, the distribution of debt and ownership, and the path by which a position can close. A generous collateral limit cannot answer those questions on its own.

For an observer, the same reasoning changes how a self-lending headline should be read. A large gross position may contain substantial internal claims; a small measured funding share may coexist with concentrated ownership or a narrow liquidation buffer. Funding attribution is a starting point for tracing dependencies. Its economic meaning emerges only after the income and recovery rights are attached to the relevant participants.

Wash lending therefore deserves analysis at both the transaction and the consolidated levels. The transaction view reveals incentives, priority and cash requirements. The consolidated view identifies the external assets that ultimately support the claims. Keeping those views consistent explains the loop's productive use of idle cash, limits the returns that can be attributed to it, and locates the questions that remain when the loop must unwind.

A External lenders and ordinary leveraged carry

When an independent lender finances the loop and the vault does not receive the borrowing interest, a collateral yield s_V gives return

$$A = s_V + m(s_V - r).$$

Ignoring fees, a lender with utilization u earns ur . If its outside option is s_0 and the borrower's alternative is unlevered vault shares, the joint improvement window is

$$\frac{s_0}{u} < r < s_V.$$

There is no requirement that lender yield be below the borrower's carry spread. The window is nonempty exactly when $us_V > s_0$. For an illustrative parameter choice, $s_V = 3.75\%$, $s_0 = 1.46\%$, $u = 80\%$ give a window from 1.825% to 3.75%. The corresponding borrower returns are

m	$r = 2.12\%$	$r = 3.00\%$
5	11.90%	7.50%
10	20.05%	11.25%
15	28.20%	15.00%

These are gross simple rates before risk, costs and compounding effects. Common denomination removes a mechanical exchange-rate mismatch at the initial mark; it does not eliminate basis, liquidity or liquidation risk.

If outside lenders instead share the same market as V , let α be the fraction of loan interest received by V , and let $T = E + Y$ still hold because all borrowing is redeposited. Then

$$c_v = \varepsilon s + \alpha r \frac{Y}{T}, \quad \sum_i e_i A_i = \varepsilon s T - (1 - \alpha) r Y.$$

The payment to outside lenders no longer cancels within the vault-holder perimeter. Market supply and the vault's own allocation must also be accounted for separately.

B TVL conventions and the financing limit

For equity e and multiplier m , collateral is $(1 + m)e$ and debt is me . A collateral-only gross TVL convention reports $(1 + m)e$; one that adds the drawn lender claim reports $(1 + 2m)e$. If all supplied liquidity is counted, including idle cash, it instead reports $(1 + m)e + S_{\mathcal{M}}$, where $S_{\mathcal{M}}$ is total market supply. None of these is net equity, which remains e for the borrower.

For $m = 15$, these first two gross measures are $16e$ and $31e$. A collateral limit of $\ell = 0.945$ allows at most $m = \ell/(1 - \ell) \approx 17.18$, but it does not provide the required loan liquidity. In the closed wash-lending system, (1) separately bounds the equity-weighted average multiplier. If every holder used $m = 15$, the vault would need $\varepsilon \leq 1/16$; a half-external allocation cannot fund that aggregate position. Gross balance-sheet amplification must therefore be reported together with the source of funds and the consolidation boundary.

C An attributed market snapshot

This appendix preserves a market-level estimate attributed to @yieldsandmore on 20 May 2026, concerning Ethena-backed lending against USDe-related collateral [5]. The source thread was not independently retrievable at preparation of this paper. The transcribed figures are used solely to illustrate the funding attribution in Section 7; they are not a verified dataset, current balances or evidence that the closed-network assumptions hold.

All amounts in USD millions. The attribution and verification status are described above.

Venue	Market	Ethena supplied	USDe+sUSDe borrowed	Lender share	Est. self- lending
Aave	Plasma USDT0	368.0	556.0	47.4%	263.3
Aave	Mantle USDT0	168.0	58.1	100.0%	58.1
Aave	Ethereum USDtb	140.0	46.7	82.5%	38.5
Aave	MegaETH USDm	550.0	398.9	91.2%	363.8
Morpho	Base Prime USDC	252.9	0.0	54.3%	0.0
Morpho	Ethereum USDtb	105.0	105.3	99.7%	105.0
Kamino	Sentora USDG	250.0	250.0	100.0%	250.0
Juplend	Bitwise USDG	251.0	251.0	100.0%	251.0
Total		2084.9	1666.0		1329.7

Using the displayed rows, attributed funding totals 1329.7 million and gross debt totals 1666.0 million. The gap is 336.3 million of debt attributed to other lenders. Products of the rounded supply shares and debt amounts need not reproduce every displayed attribution exactly. With the reported backing denominator of 4.4 billion, F/T is about 30.2%; gross collateralized debt divided by backing is about 37.9%. These are different measures, not lower and upper bounds on a verified closed-loop debt ratio.

C.1 A conditional numerical illustration

If the closed model did apply, illustrative assumptions $\varepsilon = 0.53$, $s = 9\%$, $r = 6\%$ and $\varphi = 0.30$ would give

$$c_v = 6.57\%, \quad \Delta_v = 0.57\%, \quad A(m = 6) = 9.99\%.$$

Here $u = 0.30/0.47 \approx 63.8\%$, so the aggregate funding constraint holds. Aggregate debt/equity is only $\varphi/(1 - \varphi) \approx 0.429$. If all active loopers use $m = 6$ and all other holders are passive, active looper equity must be $Y/6$, or about 7.14% of total initial equity. Calling $m = 6$ representative of every holder would be inconsistent. At $\ell = 0.915$, that local position begins liquidation after a 6.32% collateral decline and loses all initial equity at 14.29% without prior liquidation. These assumptions do not estimate Ethena’s actual carry or a loss threshold for its passive holders.

References

- [1] Larry Eisenberg and Thomas H. Noe (2001). *Systemic Risk in Financial Systems*. Management Science, 47(2), 236–249. [doi:10.1287/mnsc.47.2.236.9835](https://doi.org/10.1287/mnsc.47.2.236.9835).
- [2] Ana Fostel and John Geanakoplos (2013). *Reviewing the Leverage Cycle*. Cowles Foundation Discussion Paper No. 1918. [Cowles Foundation publication record](#).
- [3] Morpho. *Managing Bad Debt: How-to Guide*. Protocol documentation, accessed 9 September 2026. [Morpho documentation](#).
- [4] Ethena. *Rewards Mechanism*. Protocol documentation, accessed 9 September 2026. [Ethena documentation](#).
- [5] @yieldsandmore (attributed, 20 May 2026). Thread on Ethena self-lending exposure. [Original thread link](#). Verification status and use are stated in Appendix C.